CBDC Sentiment Pipeline

NLP adaptation of **CBDC (Clean Bias Direction Construction)** for tweet sentiment classification. Combines **debias_vl** word-pair projection with **CBDC text_iccv** progressive encoder training to discover and remove confound directions from BERT embeddings.

Problem

Sentiment classifiers trained on tweets can latch onto spurious correlations (topic, register, specificity) rather than genuine sentiment signal. A linear probe on BERT embeddings may learn "crypto mention = positive" instead of understanding actual sentiment.

Why vanilla CBDC fails in NLP

CBDC (CVPR 2026) assumes that "vision-language models encode semantic attributes in structurally separable subspaces" (Section 4.2). In CLIP vision, object class and spurious attributes (background, texture) are near-orthogonal, so PGD naturally discovers clean confound directions.

In BERT text space, topic and sentiment are **entangled** -- PGD-discovered directions have up to 0.94 cosine similarity with sentiment prototypes. The L_s semantic preservation loss (Section 4.4) penalizes output deviation but does not constrain the gradient direction, so PGD drifts into sentiment space when that direction reduces L_B.

Result: debias_vl (closed-form projection) improves F1 by +1.5%, but CBDC's iterative PGD refinement shows no improvement over baseline.

Method

1. **debias_vl word-pair map** -- Encode a grid of (sentiment x topic) crossed prompts. Compute the optimal debiasing projection P = P0 @ G^{-1} that preserves sentiment while removing topic confounds. Extract confound directions via SVD(I - P).

2. **CBDC text_iccv training with sentiment-orthogonal PGD** -- Use the debias_vl-discovered directions as bias anchors. Run bipolar PGD on neutral concept prompts to refine the confound axes, with **gradient projection orthogonal to sentiment prototypes** at each PGD step (`grad = grad - (grad @ S.T) @ S`). This enforces what CLIP has naturally: discovered confound directions are structurally separate from the class subspace. Iteratively train BERT's last transformer layer to produce embeddings orthogonal to the refined confound directions.

3. **Orthogonal projection** -- Apply residual projection to remove remaining confound components after encoder training.

Architecture: RN50 ↔ BERT

Component	CLIP RN50	BERT
Frozen body	ResNet conv blocks	Layers 0-10

Component	CLIP RN50	BERT
Trainable tail	attnpool (1 MHA) + c_proj	Layer 11 (1 transformer layer)
PGD perturbs	Features before attnpool	h_layer10 before layer 11
Gradient path	loss → attnpool → delta	loss → layer 11 → delta
PGD constraint	**None needed (natural orthogonality)**	**Gradient projected ⊥ sentiment prototypes**

Both tails are single-attention-layer structures. The sentiment-orthogonal PGD constraint is the key adaptation for NLP -- it compensates for the lack of structural separability in text embeddings.

Supported backbones

The encoder (`encoder.py`) resolves transformer layers dynamically and supports:

- **BERT / RoBERTa / FinBERT / DeBERTa** -- `backbone.encoder.layer`
- **DistilBERT** -- `backbone.transformer.layer`
- **GPT-2 / GPT-Neo** -- `backbone.transformer.h`
- **LLaMA / Qwen2 / Mistral** -- `backbone.model.layers`
- **CLIP text encoder** -- `backbone.text_model.encoder.layers`

> **Note:** Causal models (GPT-2, LLaMA, Qwen2) require last-token pooling and delta injection at the last non-pad position instead of CLS at position 0. This is not yet implemented.

Quick Start

```
cd project/

# Full pipeline (sentiment-orthogonal PGD enabled by default)
python run_all.py

# Resume from a phase
python run_all.py --start_phase 2

# Different backbone
python run_all.py --model bert

# Baseline only (skip CBDC training)
python run_all.py --skip_cbdc
```

Pipeline Phases

Phase	Script	Description
1	`data/embed.py`	Encode tweets with BERT

Phase	Script	Description
2	`cbdc/refine.py`	debias_vl map + CBDC text_iccv training (with sent-orthogonal PGD)
3	`pipeline/clean.py`	Orthogonal projection (all conditions)
4	`pipeline/classify.py`	Linear probe training + evaluation
5	`pipeline/evaluate.py`	Full evaluation report

Evaluation Conditions

Condition	Method	Description
B1 (raw)	None	Baseline BERT CLS embeddings
D1 (debias_vl)	Projection	debias_vl word-pair projection matrix
D2 (CBDC)	Fine-tuning	CBDC text_iccv encoder training only
D3 (CBDC+proj)	Both	CBDC encoder + residual projection
C (label-guided)	Projection	Oracle: within-class mean-shift direction

Correspondence to CBDC Paper

Paper	NLP Adaptation
CLIP text encoder (attnpool + c_proj)	BERT layer 11 (1 transformer layer)
Opposing prompt pairs ("he"/"she")	debias_vl-discovered confound directions
Neutral concept prompts ("bird")	test_text: concept descriptions
`perturb_bafa_txt_multi_ablation_lb_ls`	`_pgd_bipolar` with multi-axis L_s
`text_iccv` progressive loop	`text_iccv()` in refine.py
`debias_vl` word-pair projection	`discover_confound_map()` in refine.py
Class-specific match_loss	`S[c::n_cls] @ cls_em[c:c+1].T`
ck_loss	`l_ck(bias_a, bias_b, cls_em)`
(implicit) structurally separable subspaces	**Sentiment-orthogonal PGD gradient projection**

File Structure

```
project/
├── config.py                    # CBDCConfig (incl. sent_orthogonal_pgd
flag), MODEL_REGISTRY
├── encoder.py                   # TransformerEncoder: frozen body +
trainable tail (multi-arch)
├── losses.py                    # l_bias_contrastive,
```

```
l_semantic_preservation, l_ck
├── run_all.py                  # Pipeline orchestrator
├── data/embed.py               # Phase 1: encode and cache
├── cbdc/
│   ├── prompts.py              # All concept prompts + word-pair
definitions
│   └── refine.py               # Phase 2: debias_vl + CBDC text_iccv
├── pipeline/
│   ├── clean.py                # Phase 3: orthogonal projection
│   ├── classify.py             # Phase 4: linear probe eval
│   └── evaluate.py             # Phase 5: full evaluation report
└── cache/                      # Auto-generated intermediate files
```

Requirements

```
torch >= 2.0
transformers >= 4.30
scikit-learn
tqdm
```

Data-Grounded Prompt Design

Problem with generic topic prompts

The original DEFAULT_TOPICS used news/formal categories ("technology", "economics", "climate change"). Analysis of the TSAD tweet corpus reveals these are irrelevant -- the tweets are informal personal messages ("My arms hurt", "ooo how i love a bank holiday", "going through security already miss my baby"). The generic topics occupy a different region of BERT embedding space from the actual data, making the debias_vl projection and CBDC anchors ineffective.

Data analysis

Statistical analysis of 27,480 TSAD tweets identified two types of confounds:

Content topics (what tweets are about) -- balanced across sentiment (label entropy > 0.93):

- Daily routine: work (n=1095), school (324), sleep (396), home (718)
- Social: friends (243), family (149), people (363), party (153)
- Media: watching (339), movie (235), show (284), music (152)
- Time: today (1123), tomorrow (510), weekend (398), last night (181)

Style confounds (how tweets are written) -- the strongest spurious correlations:

Feature	neg%	neu%	pos%	Entropy
Positive emoticons (😃 😄)	15%	29%	**56%**	0.88
Ending with question mark	22%	**59%**	19%	0.87

Feature	neg%	neu%	pos%	Entropy
Multiple questions	30%	**53%**	17%	0.91
Contains URL/link	19%	**47%**	33%	0.94
Very short (<=4 words)	21%	**52%**	27%	0.93
Internet laughter (lol, haha)	16%	44%	**40%**	0.93

These style features are analogous to "background texture" in Waterbirds (CBDC paper) -- they co-occur with sentiment labels but are not causally related.

Design principle

Following debias_vl (Chuang et al. 2023), the spurious prompts define the subspace to remove. In Waterbirds, topics are "land background" and "water background". In CelebA, topics are "male" and "female". The key requirement: **topics must capture the actual confounds present in the data**.

Our prompt bank uses two template kinds:

- **Content topics** (20): `"A negative text about work."` -- analogous to Waterbirds scene categories
- **Style confounds** (12): `"A negative text written with emoticons and smiley faces."` -- analogous to Waterbirds background attributes

Both are crossed with 3 sentiment labels to form the candidate grid (32 topics × 3 sentiments = 96 candidates), preserving the same S_pairs/B_pairs structure as the original.

References

- **CBDC** (CVPR 2026) -- Clean Bias Direction Construction. PGD-based adversarial search for bias directions with semantic preservation. Section 4.2 assumes structurally separable subspaces (true in CLIP vision, not in BERT text).
- **Chuang et al. 2023** (debias_vl) -- Closed-form Orth-Cali projection P = P0(λM + I)^{-1} for debiasing vision-language representations.
- **Madry et al. 2018** -- PGD adversarial training. The sentiment-orthogonal gradient projection is analogous to projected gradient descent onto a feasible set.